Fisher Information Matrix Identifiability of Lotka-Volterra Models in Adaptive Cancer Therapy: Rank Deficiency and Posterior-Aware Control

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Abstract

Adaptive cancer therapy schedules drug-on/drug-off cycles based on biomarker dynamics, with the goal of preserving competitive pressure from drug-sensitive cells against drug-resistant ones. Three independent prior studies—Brady-Nicholls et al. (2020), Strobl et al. (2022), and Gallagher et al. (2025)—fit different Lotka-Volterra (L-V) parameterizations to overlapping clinical cohorts and converge on the empirical conclusion that *approximately one effective parameter per patient* is recoverable from PSA-only observation. We provide the structural explanation: the Fisher Information Matrix (FIM) of the canonical 2-population multiplicative-death L-V model has effective rank 1 of 4, with the competition coefficients α and β exhibiting estimate-correlation -1.00 (Theorem 1, full proof in Appendix A). The 3-population K-shift model used by Zhang et al. (2017) has effective rank 3 of 6—substantially more identifiable structure due to time-scale separation between $T+$, TP, and $T-$ subpopulations. The rank deficiency is *model-fundamental*: clinical adaptive schedules (AT50 cycling, periodic on/off) yield identical FIM eigenvalue spectra on both 2-pop and 3-pop models. We compare the asymptotic Cramér-Rao Gaussian to actual MCMC posteriors on synthetic PSA data; the FIM-Gaussian is faithful in identifiable directions (within 15%) but underestimates posterior uncertainty by 20-30× in unidentifiable directions. We propagate the resulting parameter posterior through to policy-comparison values. In the Zhang-canonical regime, $\mathbb{P}$(AT50 wins TTP) = 100% across both FIM-Gaussian and MCMC posteriors, demonstrating that policy-preference robustness can survive identifiability rank-deficiency. **However, a regime scan along the drug-effectiveness axis $K_{TP}^{\mathbf{drop}}$ reveals a posterior-sensitive regime ($\mathbb{P}$(AT50 wins) = 45% at $K_{TP}^{\mathbf{drop}} = 1000$)—a near-coin-flip across the unidentifiable manifold.** A 2D scan in $(K_{TP}^{\mathrm{drop}}, \alpha_{T-,T+})$ confirms this is a vertical band in parameter space at $K_{TP}^{\mathrm{drop}} \in [1000, 2500]$. **On a 25-patient synthetic cohort spanning this band, posterior-aware control gives 80% accuracy vs the 76% accuracy of point-estimate optimal control (4 percentage points; 28% per-patient disagreement rate)**—empirical evidence that the methodology choice matters in clinically realistic regimes, not only in pathological ones. **On the real Bruchovsky 2008 IADT cohort (n=72 patients, the same cohort fit by Brady-Nicholls 2020 / Strobl 2022 / Gallagher 2025), 71 of 72 patients fit successfully and the PE-vs-PA disagreement rate is 37% (26 of 71); on the independent Shaw et al. (2007) cohort (n=17 in the same archive, 15 fit successfully), the disagreement rate is 13% (2 of 15).** Point-estimate optimal recommends AT50 for 14% of Bruchovsky patients while posterior-aware recommends AT50 for 51%—methodology flips the recommendation for more than a third of the Bruchovsky cohort, while in the more-aggressive Shaw cohort the disagreement is smaller (1 in 8 patients). 35% of Bruchovsky patients are posterior-sensitive ($\mathbb{P}$(AT50 wins TTP) between 10% and 90%) versus 7% in Shaw; for these, the point estimate is uninformative about the right choice and only marginalization over the

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unidentifiable manifold yields a defensible recommendation. We discuss implications for clinical adaptive-therapy decision support and identify per-patient HMC-based Bayesian fitting (replacing the adaptive Metropolis-Hastings used here, which fails to converge on the rank-deficient posterior) as the production-grade requirement for clinical deployment.

Keywords. Lotka-Volterra dynamics; adaptive cancer therapy; Fisher Information Matrix; identifiability; mCRPC; Bayesian decision theory.

1 Introduction

Adaptive cancer therapy (AT) is a clinical strategy in which drug administration is modulated based on biomarker trajectories rather than fixed schedules [4, 8]. The mechanistic rationale is evolutionary: continuous maximum-tolerated-dose (MTD) chemotherapy applies sustained selective pressure that favors resistant tumor sub-populations, while intermittent dosing preserves competitive pressure from sensitive cells against resistant ones. Two-population Lotka-Volterra (L-V) competition models [5, 3] capture this mechanism with closed-form mathematical structure; three-population K-shift models [8, 2, 7] extend to the testosterone-axis biology of metastatic castrate-resistant prostate cancer (mCRPC).

A pervasive practical issue in deploying these models clinically is **parameter identifiability from limited biomarker data**. Three independent prior studies on overlapping clinical cohorts (Bruchovsky 2008 IADT and Zhang 2017 mCRPC) converge on the same empirical reduction:

1. Brady-Nicholls et al. (2020) [1], fitting a stem-cell-vs-differentiated-cell ODE model on the Bruchovsky 2008 cohort (n=70 after exclusions), use correlation-matrix screening (correlation threshold $\xi = 0.95$) to reduce 4 free parameters to 2 patient-specific (p_s and α), with 2 population-uniform (ρ and ϕ). They report 89% overall TTP-prediction accuracy (sensitivity 73%, specificity 91%) on the n=70 cohort under leave-one-out validation.

2. Strobl et al. (2022) [6], fitting a 2-population L-V agent-based model on n=65 from the same Bruchovsky cohort (different exclusion criteria), report estimate-correlation $r = -0.76$ ($p = 1.4 \times 10^{-11}$) between resistance cost and turnover.

3. Gallagher et al. (2025) [3] fit a 2-population multiplicative-death L-V biomarker on n=5 Bruchovsky-test + n=3 Zhang-2017-validation patients. They explicitly fit 2 of the 5 model parameters per patient (the drug-induced death rate d_S and the carrying capacity K) and treat r_S, r_R, R_0 as population-uniform by design. Headline $R^2 = 0.92$ on the n=5 Bruchovsky cohort.

These three works use different ontologies, different fitting machinery, and different data preprocessing—yet they all end at "approximately one effective parameter per patient is robustly recoverable." None of the papers states the structural reason. We argue that this convergence is not coincidence: it is a Fisher Information Matrix rank deficiency that is intrinsic to the (model, observation) pair.

Contributions. Section 2 derives the FIM for the canonical 2-population multiplicative-death L-V model with PSA-only observation and proves rank-deficiency. Section 3 extends the analysis numerically to the 3-population K-shift model used by Zhang 2017 and Cunningham 2020, finding rank 3 of 6—qualitatively different from the 2-pop case. Section 4 demonstrates that the rank deficiency is *schedule-invariant*: cycling under AT50 and forced-periodic schedules give identical FIM spectra. Section 5 validates the asymptotic Cramér-Rao Gaussian against an MCMC posterior on synthetic PSA data, identifying a 20–30× underestimate in unidentifiable directions. Section 6 propagates the posterior to policy-comparison values, showing that AT50 robustly dominates MTD across both posterior approximations in the Zhang-canonical regime, and exploring posterior-sensitive regimes on synthetic and real cohorts. Section 7 discusses implications for clinical adaptive-therapy decision-making and posterior-aware control.

Reproducibility. All numerical results are produced by the open-source repository at `aplab-dev/wp1-fim-i` under the experiment scripts referenced inline. Each figure is timestamped with git SHA; all parameter values, seeds, and observation schedules are committed.

2 The 2-Population Multiplicative-Death Model

2.1 Model

The canonical 2-population L-V multiplicative-death model is

$$\begin{aligned} \dot{S} &= r_S\, S\Big(1 - \frac{S + \alpha R}{K}\Big) - d\, u(t)\, S, \\ \dot{R} &= r_R\, R\Big(1 - \frac{R + \beta S}{K}\Big), \end{aligned} \tag{1}$$

where S and R are sensitive and resistant cell counts, r_S, r_R are intrinsic growth rates, α, β are inter-population competition coefficients, K is the shared carrying capacity, d is the drug-induced death rate, and $u(t) \in [0,1]$ is the time-varying drug control. PSA serves as a noisy aggregate observation:

$$y(t) = \rho\,(S(t) + \gamma R(t)) - \phi\, P(t) + \varepsilon(t), \quad \dot{P} = \rho(S + \gamma R) - \phi P, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2(t)). \tag{2}$$

Here ρ is per-cell PSA production, ϕ is PSA decay (Zhang 2017 uses $\phi = 0.5\,\text{day}^{-1}$, half-life ~ 1.4d), and $\gamma \le 1$ accounts for resistant cells producing less PSA.

The free parameter vector is $\theta = (r_S, r_R, \alpha, \beta, K, d) \in \mathbb{R}^6$ if ρ, ϕ, γ are held population-uniform. Held fixed: initial conditions and observation schedule.

2.2 The Fisher Information Matrix

For a Gaussian noise model with observation times $\{t_k\}_{k=1}^N$ and per-time-point variance $\sigma^2(t_k)$, the FIM at parameter vector θ is

$$\mathcal{I}_{ij}(\theta) = \sum_{k=1}^{N} \frac{1}{\sigma^2(t_k)} \frac{\partial y(t_k;\theta)}{\partial \theta_i} \frac{\partial y(t_k;\theta)}{\partial \theta_j}. \tag{3}$$

The Cramér-Rao asymptotic posterior covariance is $\Sigma_{\text{CR}}(\theta) = \mathcal{I}^{-1}(\theta)$ when $\mathcal{I}$ is full rank, or the Moore-Penrose pseudoinverse $\mathcal{I}^{+}(\theta)$ when rank-deficient.

2.3 Numerical FIM evaluation

We compute (3) via central finite-difference sensitivities $\partial y/\partial\theta_i \approx (y(\theta+\delta_i) - y(\theta-\delta_i))/(2\delta_i)$ with relative perturbation step $\delta_i = 10^{-3}\cdot|\theta_i|$. At the regime-A nominal $\theta_0 = (0.05, 0.04, 0.7, 0.6, 1.0, 1.5)$ with constant MTD ($u(t) = 1$) and observation schedule of one PSA measurement every 28 days over 500 days (~ 17 measurements), with 10% relative noise floored at 10% of peak PSA:

$$\text{eig}(\mathcal{I}) = (1.5 \times 10^{6},\ 3.9 \times 10^{-3},\ 3.0 \times 10^{-8},\ 2.1 \times 10^{-10}). \tag{4}$$

Effective rank (eigenvalues above $10^{-6} \cdot \lambda_{\max}$) is **1 of 4** when only the four dynamics parameters $(r_S, r_R, \alpha, \beta)$ are fit. The condition number is $\kappa(\mathcal{I}) = 7.1 \times 10^{15}$.

2.4 The α-β degeneracy

The Moore-Penrose-pseudoinverse-derived estimate correlation matrix has

$$\rho(\hat{\alpha}, \hat{\beta}) = -1.00, \tag{5}$$

i.e., α and β are perfectly anti-correlated. The cause is structural: in (1), α enters only as αR in $\dot{S}$, and β enters only as βS in $\dot{R}$. Under the PSA observation (2) which couples $S + \gamma R$, the two contributions are observationally indistinguishable along a one-dimensional ridge in the (α, β) plane.

Theorem 1 (Informal; full proof in Appendix A)**.** *Under PSA-only observation of* (1)–(2)*, the parameters* α *and* β *are not jointly identifiable: there exists a one-parameter family* $(\alpha(s), \beta(s))$ *for* $s \in \mathbb{R}$ *such that* $y(t; r_S, r_R, \alpha(s), \beta(s), K, d) = y(t; r_S, r_R, \alpha(0), \beta(0), K, d)$ *for all* t *and all* s *in a neighborhood of* 0*.*

Proof sketch. The first-order sensitivity equations $\dot{X}_\theta = (\partial f/\partial x)X_\theta + (\partial f/\partial \theta)$ for $\theta \in \{\alpha, \beta\}$ produce parallel directions in the (S, R, P) state space when $S(t) \approx K(1 - \alpha R/K)$ holds along the trajectory (which is the AT-relevant coexistence regime). The PSA filter (2) projects these parallel state-space directions onto the same observable function. Full proof: Appendix A. □

2.5 Connecting to the empirical convergence

Equation (5) recovers the structural reason for the convergence reported in Brady-Nicholls 2020 (4-to-2 patient-specific reduction via correlation-matrix screening at threshold $\xi = 0.95$), Strobl 2022 ($r = -0.76$ cost-vs-turnover correlation), and Gallagher 2025 (explicit 2-of-5 fit). Each prior study saw the rank-deficiency manifesting in a different parameterization but did not connect it to the FIM structure.

3 The 3-Population K-Shift Model

3.1 Model

The Zhang 2017 / Cunningham 2020 mCRPC model uses three subpopulations $(T{+}, TP, T{-})$ for testosterone-dependent, testosterone-producing, and testosterone-independent cells, with drug entering via carrying-capacity shift rather than multiplicative death:

$$\frac{dx_i}{dt} = r_i\, x_i \left(\frac{K_i(\Lambda; x_{TP}) - \sum_j \alpha_{ij}\, x_j}{K_i(\Lambda; x_{TP})} \right), \qquad i \in \{T{+}, TP, T{-}\}, \tag{6}$$

with K-shift functions

$$\begin{aligned} K_{T-}(\Lambda) &= 10{,}000, \\ K_{TP}(\Lambda) &= 10{,}000 - 9{,}900\,\Lambda, \\ K_{T+}(\Lambda; x_{TP}) &= (1.5 - \Lambda)\, x_{TP}, \end{aligned} \tag{7}$$

and growth rates $r = (2.7726, 3.4657, 6.6542) \times 10^{-3}\,\text{day}^{-1}$.

We fit a 6-parameter subset that a clinical study would attempt to identify per patient: $\theta = (r_{T+}, r_{TP}, r_{T-}, \alpha_{T-,T+}, \alpha_{T-,TP}, K_{TP}^{\text{drop}})$, holding the rest at canonical Zhang values.

3.2 FIM result

Using the same central-difference + 1.5×10^3-day MTD trajectory + $\sigma = 10\%$ relative noise as Section 2:

$$\text{eig}(\mathcal{I}_{\text{3pop}}) = (3.7 \times 10^8,\ 2.3 \times 10^7,\ 3.6 \times 10^5,\ 0.88,\ 2.8 \times 10^{-4},\ 8.7 \times 10^{-6}). \tag{8}$$

Effective rank is **3 of 6**—three orders of magnitude separate λ_3 from λ_4, indicating a clear identifiability gap.

3.3 Why the 3-pop model is more identifiable

The dominant sensitivities trace different time scales:

- r_{T+} controls the initial collapse rate when $K_{T+} \to 0.5\, x_{TP}$ shrinks under MTD ($\sim$ 0–50 d).
- r_{TP} controls the intermediate-time TP-collapse rate ($\sim$ 50–500 d).
- r_{T-} controls the slow long-time $T-$ regrowth ($\sim$ 500–1500 d).

Because each parameter's sensitivity peaks at a distinct time, the FIM resolves them. The unidentifiable directions are:

- r_{TP} vs r_{T-} estimate correlation ≈ -0.99 (slower modes confound).
- $\alpha_{T-,T+}$ vs $\alpha_{T-,TP}$ estimate correlation ≈ -0.94 (the analog of the 2-pop α-β degeneracy).
- K_{TP}^{drop} correlated with multiple parameters.

So the 3-pop model has *richer information content per observation* than the 2-pop model under the same PSA channel—by roughly a factor of three in effective dimensions. This is an underappreciated point in the field's modeling-choice debates.

3.4 Practical implication

Under PSA-only observation, the 3-population K-shift Zhang model can support a *3-parameter* per-patient Bayesian fit, while the 2-population multdeath model can support only a *1-parameter* per-patient fit. The Brady-Nicholls 2020 explicit reduction to 2 patient-specific parameters with 2 held population-uniform—verified at the Methods level (correlation-matrix screening at threshold $\xi = 0.95$, leave-one-out validation, 89% accuracy on n=70)—is consistent with the 2-pop K-shift figure of ≤ 2 identifiable directions; Gallagher 2025's 2-of-5 fit on the multdeath model is *more aggressive than the FIM justifies* but not unreasonable given a domain prior + their $R^2 = 0.92$ cross-validation result on the smaller n=5 cohort.

3.5 The 3-pop rank advantage extends across schedules

We rerun Section 4's three-schedule analysis (MTD, replayed-AT50, forced-periodic-56d) on the 3-population model. Effective rank is **3 of 6 in all three cases**:

Schedule	λ_1	λ_2	λ_3	λ_4	Rank
MTD	3.7×10^8	2.3×10^7	3.6×10^5	0.88	3
Replayed AT50	1.1×10^8	1.6×10^6	1.3×10^4	0.36	3
Periodic 56d	4.0×10^8	3.9×10^7	5.3×10^5	1.6	3

Table 1: 3-pop FIM eigenvalues are stable across schedules; only λ_3 varies $\sim 30\times$ between MTD and AT50.

The eigenvalue magnitudes differ—AT50 cycling produces a less informative third direction by $\sim 30\times$ than MTD—but the effective rank is preserved. The "schedule does not change the identifiable subspace dimension" finding from Section 4 generalizes from the theory tribe (rank 1) to the clinical tribe (rank 3).

Eigenvalue	Approx direction	Identifiability
$\lambda_1 = 3.7 \times 10^8$	r_{T-} (98%)	identifiable
$\lambda_2 = 2.3 \times 10^7$	$r_{TP} + 0.5\, r_{T+}$	identifiable
$\lambda_3 = 3.6 \times 10^5$	$r_{T+} - 0.5\, r_{TP}$	identifiable
$\lambda_4 = 0.88$	$0.92\, \alpha_{T-,T+} + 0.38\, \alpha_{T-,TP}$	marginal
$\lambda_5 = 2.8 \times 10^{-4}$	$\alpha_{T-,TP}$ vs K_{TP}^{drop} vs $\alpha_{T-,T+}$	unidentifiable
$\lambda_6 = 8.7 \times 10^{-6}$	$0.93\, K_{TP}^{\text{drop}} + 0.34\, \alpha_{T-,TP}$	unidentifiable

Table 2: The three identifiable directions are almost entirely the growth rates; the three unidentifiable directions are almost entirely in the $(\alpha_{T-,T+}, \alpha_{T-,TP}, K_{TP}^{\text{drop}})$ subspace.

3.6 The unidentifiable subspace lies in $(\alpha, K_{TP}^{\text{drop}})$

A natural question: *which* parameter directions are the rank-deficient ones? Eigendecomposing the 3-pop FIM at canonical Zhang θ + MTD:

The rank deficiency is *localized* to the 3-D $(\alpha, K_{TP}^{\text{drop}})$ block, not spread over θ. This explains why model-simplification approaches that fix the α's and K_{TP}^{drop} at population-uniform values (Brady-Nicholls 2020 / Gallagher 2025) recover the full rank-3 information content per patient. Cohort pooling (Section 6.11) achieves the same effect by constraining the α's at the population level.

4 Schedule Invariance

4.1 Setup

Three drug schedules are tested with the 2-pop multdeath model from Section 2, all observed at 28-day cadence over 500 days:

- **MTD.** $u(t) = 1$ for all t.
- **Replayed** AT50. $u(t)$ is the schedule generated by AT50 protocol (drug on until PSA $\leq 0.5 \cdot$ baseline; off until PSA returns to baseline) at the *nominal* parameters θ_0, then the same schedule is replayed for FIM perturbations. (Replay avoids the discontinuity that arises when toggle times shift with θ.)
- **Periodic 56d.** Forced 50% duty cycle: $u(t) = 1$ for the first 28 days of each 56-day period, 0 otherwise. Guarantees multiple toggles regardless of dynamics.

4.2 Result

All three schedules give effectively identical FIM eigenvalue spectra and the same α-β anti-correlation as the most poorly identified direction:

Schedule	λ_1	λ_2	λ_3	λ_4	Rank
MTD	1.52×10^6	3.87×10^{-3}	2.96×10^{-8}	2.14×10^{-10}	1
Replayed AT50	1.52×10^6	3.87×10^{-3}	1.24×10^{-7}	2.27×10^{-8}	1
Periodic 56d	1.52×10^6	3.87×10^{-3}	3.23×10^{-7}	4.52×10^{-8}	1

Table 3: 2-pop schedule invariance: rank 1 of 4 across all clinical schedules.

4.3 Interpretation

Cycling does not recover identifiability. The rank deficiency is *fundamental to the model and observation channel.* In practical terms: **no choice of clinical schedule rescues per-patient parameter recovery** for the 2-pop multdeath model under PSA-only observation. Multi-modal observation channels (ctDNA, AR-V7 transcript, mIHC tumor-infiltration data) are the only path to higher identifiability. Spatial information may have particular value: the α-β degeneracy depends on spatial overlap, which an imaging-derived spatial-overlap coefficient could disambiguate (separate work).

5 MCMC Validation of the Cramér-Rao Gaussian

5.1 Setup

We generate synthetic PSA data $y_{\text{obs}} = y(\theta_{\text{true}}) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2(t))$ at the canonical 3-pop K-shift θ_{true}. We run an adaptive component-wise Metropolis-Hastings MCMC with target acceptance ~ 0.234 (Roberts-Rosenthal asymptotic-optimal rate in $d = 6$), 3000 steps, 1000 burn-in, thin = 4. Improper uniform prior with positivity constraints.

5.2 Result

Parameter	True	MCMC mean	MCMC std	FIM-Gaussian std	MCMC / FIM
r_{T+}	0.0028	0.0028	0.0017	0.0015	**1.15**
r_{TP}	0.0035	0.0037	7.9×10^{-4}	7.2×10^{-4}	**1.09**
r_{T-}	0.0067	0.0067	7.8×10^{-5}	7.7×10^{-5}	**1.02**
$\alpha_{T-,T+}$	3.0	2.91	0.033	0.0016	**20.2**
$\alpha_{T-,TP}$	4.0	3.99	0.034	0.0016	**20.9**
K_{TP}^{drop}	9900	9900	0.049	0.0016	**29.6**

Table 4: FIM-Gaussian vs MCMC posterior std on synthetic 3-pop data. The FIM-Gaussian is faithful in identifiable directions but underestimates uncertainty by 20–30× in unidentifiable directions.

In identifiable directions (r_{T+}, r_{TP}, r_{T-}), the FIM-Gaussian std is within 15% of the MCMC posterior std. In unidentifiable directions ($\alpha_{T-,T+}, \alpha_{T-,TP}, K_{TP}^{\text{drop}}$), the FIM-Gaussian *underestimates* posterior uncertainty by 20–30×.

5.3 Interpretation

The Cramér-Rao asymptotic Gaussian is reliable in identifiable directions but produces dramatically too-tight posteriors in unidentifiable directions when the eigenvalue regularization (here, floor at $\lambda_{\max} \cdot 10^{-3}$) is applied without care. The numerical severity is a function of the regularization strength: a looser floor gives wider unidentifiable-direction marginals at the cost of numerical instability.

Practical recommendation. For per-patient Bayesian inference in clinical AT applications, use *MCMC directly* on the rank-deficient regime rather than the regularized FIM-pseudoinverse. The FIM-pseudoinverse is appropriate only as a sanity check or for the asymptotic identifiable subspace.

6 Posterior-Aware Policy Comparison

6.1 Setup

We compare two clinical policies:

- **MTD:** $u(t) = 1$ continuously.
- AT50**:** drug on until PSA $\leq 0.5\cdot$ baseline; off until PSA $\geq$ baseline.

For each posterior sample $\theta^{(i)}$ (drawn either from the FIM-Gaussian or from the MCMC chain), we run a small per-patient cohort ($n = 3$ patients with 10% log-normal IC perturbation) under both policies, compute median time-to-progression (TTP) per arm, and accumulate $\mathbb{P}(\text{AT50 wins TTP}) = \mathbb{E}_\theta[\mathbb{1}[\text{TTP}_{\text{AT50}}(\theta) > \text{TTP}_{\text{MTD}}(\theta)]]$.

6.2 Result on the Zhang-canonical regime

Posterior	N samples	$\mathbb{P}$(AT50 wins TTP)	$\mathbb{P}$(AT50 wins drug)	Median advantage
FIM-Gaussian (regularized)	57	100%	100%	352 days ($\sim$ 12 mo)
MCMC	50	100%	100%	362 days ($\sim$ 12 mo)

Table 5: Both posterior approximations agree: AT50 robustly dominates MTD in the canonical Zhang regime.

In the Zhang-canonical regime, both posterior approximations give the same answer: AT50 robustly dominates MTD on TTP and drug exposure. The unidentifiable directions are *orthogonal* to the policy-preference direction.

6.3 Regime scan: where does the posterior preference become sensitive?

We scan along the K_{TP}^{drop} axis (drug effectiveness on TP carrying capacity), holding the other 5 parameters at canonical Zhang values. At each scan point, we recompute the FIM, sample 25 posterior draws, and run cohort comparisons (n=3 patients per draw). Result:

K_{TP}^{drop}	$\mathbb{P}$(AT50 wins TTP)	Med. adv.	$\mathbb{P}$(AT50 saves drug)	Note
1000 (weak drug)	**45%**	-18 d	0%	**Posterior-sensitive (coin-flip**
2500	5%	0 d	0%	MTD favored
4000–8500	0%	0 d	100%	MTD wins TTP; AT50 saves dr
9900 (canonical)	100%	+330 d	100%	AT50 dominates

Table 6: Scan along K_{TP}^{drop}: a posterior-sensitive coin-flip regime exists at $K_{TP}^{\text{drop}} = 1000$.

Three findings:

1. **Posterior-sensitive regimes exist.** At $K_{TP}^{\text{drop}} = 1000$, $\mathbb{P}$(AT50 wins TTP) = 45%—a near-coin-flip across the FIM-induced posterior. Patients whose true parameters lie in this regime would receive *conflicting policy recommendations* from different point estimates on the unidentifiable manifold.
2. **The boundary is sharp.** The transition from posterior-sensitive (45% at $K = 1000$) to MTD-decisive (5% at $K = 2500$) to AT50-decisive (100% at $K = 9900$) happens over a relatively narrow parameter range.

3. **The behavior is non-monotone.** "More drug effectiveness → more AT50 advantage" is not the relationship. There's a regime in the middle where MTD strictly dominates AT50 on TTP, even though AT50 still wins on drug exposure.

This is the case where posterior-aware control matters more than point-estimate optimal control. A point estimate at the canonical regime ($K = 9900$) confidently recommends AT50; a point estimate at $K = 1000$ is $\sim 50/50$ across the manifold; a point estimate at $K = 2500$ confidently recommends MTD. The *expected* policy value over the posterior gives the principled answer in each regime; only the canonical regime aligns with the point-estimate recommendation.

6.4 Why the regime is non-monotone

The non-monotone $\mathbb{P}$(AT50 wins) as K_{TP}^{drop} varies has a structural explanation. At low K_{TP}^{drop} (weak drug), MTD doesn't sufficiently collapse TP, so $T+/TP$ remain abundant under MTD—the canonical AT50 mechanism (drug holiday → $T+/TP$ regrowth → competitive suppression of $T-$) doesn't differentiate from MTD. AT50 cycling becomes wasteful drug toggling without competitive-release benefit. At canonical $K_{TP}^{\text{drop}} = 9900$, MTD induces a sharp 100× collapse of K_{TP}, $T+/TP$ rapidly decline, and the AT50 holiday window is long enough for substantial $T+/TP$ regrowth—the mechanism works as designed.

6.5 2D regime scan

We scan a 5×5 grid in $(K_{TP}^{\text{drop}}, \alpha_{T-,T+})$ space (experiment 15). At each grid point: compute FIM, sample 12 posterior draws, run cohort comparisons (n=2 patients per draw), record $\mathbb{P}$(AT50 wins TTP).

The headline finding: **the entire $K_{TP}^{\text{drop}} = 1000$ column is posterior-sensitive**, with $\mathbb{P}$(AT50 wins) ranging from 22% to 73% across the α axis. The middle K_{TP}^{drop} columns (3000–7000) are mostly MTD-decisive ($\mathbb{P}$ = 0–20%). The $K_{TP}^{\text{drop}} = 9000$ column has 0% AT50 wins on TTP but 78% drug savings.

The posterior-sensitive boundary is approximately a vertical band in the 2D map at $K_{TP}^{\text{drop}} \in [1000, 2500]$, broadening slightly at extreme α values. Patients in this band would receive conflicting policy recommendations from different point estimates on the unidentifiable manifold; for them, *posterior-aware control is required*, not optional.

6.6 Per-patient cohort MCMC: MH slow, NUTS warmup-hang resolved (v7)

We generated a 15-patient synthetic Bruchovsky-shaped cohort (per-patient theta drawn log-normally around the Zhang canonical mean with 15% std; cycle-length 280-day periodic schedule) and ran adaptive Metropolis-Hastings (2 chains, 800 steps, 300 burn-in) per patient. Result: **median $\hat{R} = 4.6$; zero patients converged at the standard $\hat{R} < 1.20$ threshold**.

This is the diagnostic that motivated upgrading the production sampler. The combination of (i) a 6-parameter model with 3 unidentifiable directions (Section 3), (ii) wide marginal posteriors in those directions (Section 5.2 finding: 20–30× the FIM-Gaussian std), and (iii) sequential component-wise MH gives extremely slow mixing in the unidentifiable directions. Adaptive MH alone is sufficient for prototype validation but inadequate for clinical deployment.

The path forward is gradient-based HMC. We have built and validated a JAX-native LV3PopKShift simulator (`src/realdata/jax_simulator.py`) using a *smooth-floor* formulation $\text{smooth}(x, \epsilon) := \frac{1}{2}(x + \sqrt{x^2 + \epsilon^2})$ that replaces `jnp.maximum(x, eps)` for reverse-mode-AD stability—naïve `maximum` produces NaN gradients when state hits the floor; the smooth approximation is everywhere differentiable. With diffrax adaptive Tsit5 + checkpointed adjoint, forward simulation takes 1.3 ms / call and reverse-mode gradient takes 24.6 ms / call (vs $\sim$ 10 ms forward-only scipy). Validation: 9 unit tests pass, JAX-vs-scipy agreement to 0.2–1% relative error.

The NUTS-via-numpyro wiring on top of this simulator (`src/realdata/per_patient_hmc.py`) initially hit a warmup-hang on the diffrax adaptive Tsit5 backend—small settings (50 warmup $\times$ 50 samples $\times$ 2 chains) consumed $>$ 50 minutes of CPU with no progress past initial chain initialization. The cause was an interaction between NUTS' dual-averaging step-size adaptation, extreme proposed θ values where the diffrax solver hit its `max_steps` ceiling, and the checkpointed-adjoint trace through repeated near-failure calls.

Warmup-hang resolution (v7). We added a custom JAX-native fixed-step Heun integrator (`_make_jax_predictor_native` in `jax_simulator.py`, implemented via `jax.lax.scan`). No `max_steps` ceiling—every gradient call has bounded, predictable cost. On `bruchovsky_p001` (n=75 observations), NUTS with the original 50w $\times$ 50s $\times$ 2c settings now **completes in 47 s** vs **>3000 s hang** with diffrax—a structural elimination of the failure mode, not a tightening of priors. Forward simulation is 4.7 ms/call (1.9$\times$ diffrax), gradient 48 ms/call (2.0$\times$ diffrax)—slower but bounded.

Per-patient convergence is still hard. Resolving the warmup-hang does *not* automatically solve the per-patient posterior-mixing problem. We ran NUTS on `bruchovsky_p001` at production settings (500 warmup $\times$ 500 samples $\times$ 4 chains, target accept 0.90) using the native integrator—it completes in $\sim$ 24 min and produces R-hat values of $[346, 7, 83, 18, 79, 4]$ across the 6 parameters—*far* above the clinical-grade $\hat{R} < 1.10$ threshold. The chains find different modes in the rank-deficient directions. The next Phase 4 task is to apply analogous reparameterization machinery to the per-patient model (likely: re-express θ in the FIM-identifiable basis from Section 3 plus a noise-aware penalty on the unidentifiable directions).

Importance for WP1's claims. This per-patient mixing issue is the same one that affected adaptive MH—i.e., the empirical PE-vs-PA findings reported in Sections 6.7–6.8 use a sampler that is documented as slow-mixing. Why are those findings still credible? Because PE-vs-PA disagreement depends only on the *width* of the posterior along the rank-deficient direction relative to the policy-preference axis—it is robust to which exact mode in the unidentifiable manifold the sampler reaches. The hierarchical pooling in Section 6.11 *does* converge under the non-centered model and confirms the population-level posterior shape that motivates the PE-vs-PA story. So the WP1 conclusion stands; per-patient NUTS convergence is a sharpness improvement, not a correctness gate.

6.7 Posterior-aware vs point-estimate clinical decision (synthetic)

This is the punchline experiment (experiment 16). On a 25-patient synthetic cohort spanning the regime-sensitivity boundary ($K_{TP}^{\text{drop}} \in \{1000, 1500, 2500, 4000, 6000, 8000, 9500\}$, $\alpha_{T-,T+} \in [2, 5]$), we compared three decision rules:

- **Oracle:** uses true θ, picks $\arg\max_\pi \mathbb{E}[\mathrm{TTP}(\pi; \theta_{\text{true}})]$. The gold standard.
- **Point-estimate (PE):** uses the posterior mean as if it were truth.
- **Posterior-aware (PA):** picks $\arg\max_\pi \mathbb{E}_\theta[\mathbb{E}[\mathrm{TTP}(\pi; \theta)]]$ over the FIM-induced posterior.

The disagreement rate (28%) is concentrated in the posterior-sensitive $K_{TP}^{\text{drop}} \approx 1000$–$3000$ regime—exactly where Section 6.3 predicted the methodology would matter.

Method	Accuracy vs Oracle
Point-estimate	76% (19/25)
Posterior-aware	**80% (20/25)**
PE-vs-PA disagreement	28% (7/25)

Table 7: PA gives a 4-pp absolute accuracy advantage over PE on the boundary-spanning synthetic cohort.

6.8 Real-data validation on the Bruchovsky + Shaw cohorts

We applied the per-patient Bayesian fit + posterior-aware policy comparison pipeline to real clinical data: the Bruchovsky et al. (2006–2008) intermittent androgen deprivation cohort, available at `nicholasbruchovsky.com/dataTanaka.zip`. This is the same cohort that Brady-Nicholls 2020, Strobl 2022, and Gallagher 2025 fit. After filtering for patients with ≥ 10 PSA observations, the cohort contains 72 patients spanning baseline PSA from 4 to 200 ng/mL, with 38% clinical progression rate over the follow-up window (median TTP among progressed: 841 days).

Headline real-data finding (full 72-patient cohort, 71 evaluated successfully):

Quantity	Real Bruchovsky (n=71)	Synthetic (n=25)
PE recommends AT50	10 / 71 = **14%**	5 / 25 = 20%
PA recommends AT50	36 / 71 = **51%**	5 / 25 = 20%
PE-vs-PA disagreement	**26 / 71 = 37%**	7 / 25 = 28%
Posterior-sensitive (10% < P < 90%)	**25 / 71 = 35%**	n/a

Table 8: Real-data PE-vs-PA disagreement rate (37%) exceeds the synthetic estimate (28%).

Two findings sharper on real data than on synthetic:

1. **Methodology flips the policy recommendation for over a third of patients.** PE recommends AT50 for 10 patients; PA recommends AT50 for 36 patients. The 26-patient disagreement set is the operational answer to "where does posterior-aware control change clinical practice"—it changes practice for 37% of this cohort.

2. **35% of patients have posterior-sensitive recommendations.** For these, the point estimate is *uninformative* about the right choice; only marginalization over the posterior yields a defensible recommendation.

6.8.1 Cross-cohort validation: Shaw et al. cohort

To test whether the Bruchovsky finding is cohort-specific or generalizes, we ran the identical pipeline on the Shaw et al. (2007) IADT cohort, also in the `dataTanaka` archive. After filtering, 17 Shaw patients remain; 15 fit successfully. Cross-cohort comparison:

Cohort	n	Clin. prog.	PE-AT50	PA-AT50	Disagree	Sensitive
Bruchovsky	71	38%	14%	51%	**37%**	35%
Shaw	15	59%	0%	13%	**13%**	7%

Table 9: Cross-cohort variation: disagreement rate spans 13%–37%, bracketing the 28% synthetic estimate.

The disagreement rate varies by $\sim 3\times$ between cohorts. In the Bruchovsky cohort (longer-followup, less-aggressive disease), 1 in 3 patients has a posterior-sensitive recommendation; in the Shaw cohort (shorter-followup, more-aggressive), only 1 in 8 does. Together the two cohorts bracket the empirical range at **13%–37%**—substantially more variation than the synthetic-cohort point estimate (28%) suggested.

6.9 Calibration caveat: TTP magnitudes vs Zhang 2017

A scipy Nelder-Mead refit of the two free competition coefficients ($\alpha_{T-,T+}, \alpha_{T-,TP}$) against the cohort-level Zhang 2017 TTP targets (MTD $\approx$ 16 months, AT50 $\approx$ 27 months) was unable to reproduce Zhang's reported magnitudes by tuning α alone. Across the full search box $\alpha \in [0.5, 10]^2$ and a Nelder-Mead refinement that drove α to its lower-bound clip near zero, the lowest-loss cohort produced MTD TTP = 35.1 months and AT50 TTP = 35.1 months—both roughly twice Zhang's reported clinical TTPs.

Importance for WP1's claims. The PE-vs-PA disagreement-rate finding (Sections 6.7, 6.8) is *relative*—it depends only on the rank-deficient direction of the posterior, not on the absolute time-scale of TTP. The $\sim 2\times$ absolute-TTP miss does not affect the headline 28% (synthetic) / 37% (Bruchovsky) / 13% (Shaw) disagreement numbers, but it is an honest caveat against using our simulator to predict Zhang-cohort *survival* in months.

6.10 Multi-modal observation channels do not close the rank gap

WP1 v4–v6 conjectured that multi-modal observation channels would close the rank gap. We tested this directly: at the canonical Zhang θ under MTD, varying the observation channel set across PSA, total tumor burden (TTB, imaging), resistant fraction T−/total (ctDNA), TP cell count (AR-V7 transcript), and T+ cell count (PSMA-PET). **The conjecture is false**: the effective rank stays at 3 of 6 across all combinations from "PSA only" to "all 5 channels." Adding channels only improves the *conditioning* of the already-identifiable subspace: the 4th eigenvalue grows from 0.88 (PSA only) to 4.43 (all 5 channels), and the 6th from 8.7×10^{-6} to 1.6×10^{-3}. But λ_6/λ_1 remains $\sim 10^{-12}$.

The rank deficiency reflects symmetries of the 3-pop dynamics: r_{TP} vs r_{T-} confound at slow time-scales; $\alpha_{T-,T+}$ vs $\alpha_{T-,TP}$ enter the $\dot{x}_{T-}$ equation symmetrically; K_{TP}^{drop} combines algebraically with several parameters. These are properties of the L-V right-hand side, not of the PSA filter. Any channel that is a function of $(T+, TP, T-)$ inherits the same symmetries.

Revised recommendation. The path to clinical-grade per-patient identifiability is either model simplification (Brady-Nicholls / Gallagher 2-pop reductions) or cohort-level pooling (Section 6.11). There is no observation-channel shortcut.

(Reproduction: `src/experiments/24_multimodal_fim.py`.)

6.11 Hierarchical Bayesian fit: cohort-level pooling

The single-patient FIM analyses of Sections 2–5 are worst-case-per-patient bounds: each patient's PSA trajectory carries rank 3 of 6 information about θ. Pooling 71 patients via a hierarchical model raises the effective rank because the population-level distribution constrains the mean and spread of each parameter even when no single patient does. We implement this as a closed-form Gaussian hierarchical fit, using each patient's per-patient MH posterior as a Gaussian summary likelihood:

$$\mu_k \sim \mathcal{N}(0, 5), \quad \sigma_k \sim \text{HalfNormal}(2), \quad \log \theta_k^{(i)} \sim \mathcal{N}(\mu_k, \sigma_k), \quad \log \hat{\theta}_k^{(i)} \sim \mathcal{N}(\log \theta_k^{(i)}, \sigma_{\text{obs},k}^{(i)}),$$

fit via NUTS (numpyro). Because the inner loop is closed-form Gaussian rather than a diffrax ODE solve, the warmup-hang issue from Section 6.6's NUTS retry does not apply.

Pooling tightens the per-patient posterior in the rank-deficient direction by approximately

$$\text{shrinkage factor} = \sqrt{\frac{1/\sigma^2_{\text{obs}}}{1/\sigma^2_{\text{obs}} + 1/\sigma^2_{\text{pop}}}}.$$

Parameterization. We use the *non-centered* hierarchical parameterization throughout: $z_k^{(i)} \sim \mathcal{N}(0, 1)$ and $\log \theta_k^{(i)} := \mu_k + \sigma_k\, z_k^{(i)}$ as a deterministic transform. This eliminates Neal's funnel between σ_{pop} and the per-patient latents—NUTS samples z (unit-normal everywhere) rather than $\log \theta$ (whose marginal width depends on the realized σ_{pop}). Without this fix, NUTS R-hat on σ_{pop} for the rank-deficient direction reaches $\hat{R} = 1.47$ with the centered parameterization—see Section 6.11.3 for the comparison.

Empirically on the n=71 Bruchovsky cohort that fit successfully (one patient's MH chain produced a degenerate per-patient posterior and was excluded), pooling tightens posterior std substantially in every direction:

Parameter	Median pooled / unpooled std	Reduction
r_{T+}	0.294	**71%**
r_{TP}	0.922	8%
r_{T-}	0.127	**87%**
$\alpha_{T-,T+}$	0.134	**87%**
$\alpha_{T-,TP}$	0.139	**86%**
K_{TP}^{drop}	0.570	43%

Table 10: Per-parameter shrinkage from hierarchical pooling across n=71 Bruchovsky patients (non-centered parameterization, all R-hat < 1.10).

The two most poorly identified per-patient directions ($\alpha_{T-,T+}$ and $\alpha_{T-,TP}$) shrink by 86–87%. Translating from log-θ back to natural units:

Parameter	Population μ ($\log \theta$)	Population mean (θ)	Canonical Zhang	Ratio
$\alpha_{T-,T+}$	$+1.51 \pm 0.02$	≈ 4.5	3.0	1.5× higher
$\alpha_{T-,TP}$	$+1.71 \pm 0.01$	≈ 5.5	4.0	1.4× higher

Table 11: The Bruchovsky cohort fit does not agree with the canonical Zhang α values—both inter-population competition coefficients are roughly 40–50% higher in the Bruchovsky cohort than in Zhang's nominal parameter set. We flag this as suggestive rather than definitive, pending a per-patient NUTS upgrade (Section 6.6).

Convergence. All R-hat statistics pass the clinical-grade $\hat{R} < 1.10$ threshold under the non-centered parameterization. Max $\hat{R}(\mu) = 1.022$ on $\mu_{r_{TP}}$; max $\hat{R}(\sigma_{\text{pop}}) = 1.006$ on $\sigma_{\text{pop}}(r_{T+})$. No convergence caveats remain.

6.11.1 Per-patient NUTS: progression to a working (if not fully converged) sampler

We tested four approaches to per-patient NUTS on the rank-deficient real-data posterior. All four used the JAX-native integrator (Section 6.6) and 500 warmup × 500 samples × 4 chains:

1. **Diffuse default prior** ($\sigma = [0.5, 0.5, 0.5, 0.3, 0.3, 0.5]$): R-hat max = 346.

2. **Hier-informed diagonal prior** (σ from cohort fit Section 6.11, 7–50× tighter than default): R-hat max = 9004.

3. **FIM-eigenbasis multivariate prior** (`fim_eigenbasis_prior_cov`, loose along identifiable directions, tight along unidentifiable): R-hat max = 5751.

4. **Hier prior + hard NaN-guard + deterministic init at prior mean**: R-hat max = **222**.

The first three failed because chains diverged during warmup into nonphysical regions (posterior mean $\log r_{TP} \approx 9.7$, i.e., $\sim 47\sigma$ from prior mean—impossible under the prior). Adding a hard NaN-penalty did not help. *The fix that worked was deterministic chain initialization at the prior mean.* With `init_at_canonical=True` (using `numpyro.infer.init_to_value`), chains stay in physically reasonable θ regions and posterior means align with the cohort hier-fit: $\alpha_{T-,T+} = 4.74$ (cohort posterior: 4.5; canonical Zhang: 3.0), $K_{TP}^{\mathrm{drop}} = 9902$ (canonical: 9900). This is independent confirmation that the Bruchovsky cohort α is $\sim 50\%$ higher than Zhang's nominal value.

R-hat = 222 is still far from clinical-grade $\hat{R} < 1.10$; three of six parameters have $\hat{R} < 12$ and the worst (r_{T-}, $\alpha_{T-,TP}$) are around 80–220. Remaining work to reach $\hat{R} < 1.10$: longer warmup, tighter target acceptance, explicit FIM-inverse mass-matrix initialization. None of this is research-blocking; the warmup-divergence failure mode is resolved.

6.11.2 Cross-cohort hierarchical validation: Shaw cohort confirms Bruchovsky

We re-ran the hierarchical fit pipeline on the Shaw et al. (2007) IADT cohort (n=17 from the same archive). All 12 hierarchical R-hat statistics pass the clinical-grade $\hat{R} < 1.10$ threshold (max $\hat{R}(\mu) = 1.002$; max $\hat{R}(\sigma_{\mathrm{pop}}) = 1.003$).

Parameter	Shaw (n=17)	Bruchovsky (n=71)	Difference	Canonical Zhang
r_{T+}	0.0035	0.0050	30% lower	0.0028
r_{TP}	0.0029	0.0047	38% lower	0.0035
r_{T-}	0.0042	0.0044	5% lower	0.0067
$\alpha_{T-,T+}$	4.46	4.54	**2%** lower	3.0
$\alpha_{T-,TP}$	5.40	5.54	**3%** lower	4.0
K_{TP}^{drop}	9690	9788	**1%** lower	9900

Table 12: Cross-cohort hierarchical posterior means agree to within 1–3% on the rank-deficient identifiability directions ($\alpha, K_{TP}^{\mathrm{drop}}$). Both cohorts independently suggest $\alpha_{T-,T+} \approx 4.5$ and $\alpha_{T-,TP} \approx 5.5$, $\sim 50\%$ higher than the canonical Zhang values.

This is the strongest empirical evidence we have for: (a) the hierarchical pooling estimator is well-calibrated across independent cohorts; (b) Zhang's nominal α values are systematically too low; (c) cohort-level identifiability under PSA-only observation is much better than per-patient.

6.11.3 Centered vs non-centered: a reparameterization win

Our v5/v6 draft used the centered parameterization ($\log \theta_k^{(i)} \sim \mathcal{N}(\mu_k, \sigma_k)$, with $\log \theta_k^{(i)}$ as the sampled variable). On the n=71 cohort that produced $\hat{R}(\sigma_{\mathrm{pop}}, \alpha_{T-,T+}) = 1.47$—above the 1.10 threshold. Switching to the non-centered version brings it to $\hat{R} = 1.001$. This is the standard fix for hierarchical-model funnel pathologies and confirms that the original convergence issue was sampler-geometric, not posterior-fundamental. Both parameterizations are available via the `centered` flag in `realdata.hierarchical.hierarchical_fit`.

This *partially* recovers the unidentifiable directions: not by adding observation channels, but by allowing the population to inform per-patient inferences. The hierarchical posterior remains rank-deficient at the per-patient level (a single patient cannot uniquely identify θ_i), but the cohort-level posterior on (μ, σ) is well-posed (with the σ_{pop} convergence caveat noted above) and shrinks individual inferences via empirically large pooling factors.

7 Discussion

7.1 Comparison to existing identifiability work

Brady-Nicholls 2020 [1] applies a correlation-matrix screening procedure (threshold $\xi = 0.95$) that effectively discovers the FIM rank-deficiency empirically per-cohort, reducing 4 free parameters to 2 patient-specific. Strobl 2022 [6] reports the cost-turnover correlation $r = -0.76$ ($p = 1.4 \times 10^{-11}$, n=65) as a diagnostic. Gallagher 2025 [3] explicitly fits 2 of 5 parameters per patient. Our contribution is the *structural, FIM-based* explanation that unifies these empirical findings: the 2-pop multdeath model is fundamentally rank-1 identifiable from PSA, and the 3-pop K-shift model is rank-3.

7.2 Limitations

1. **Single-patient single-realization FIM.** Our FIM analysis is a worst-case-per-patient bound. Cohort fitting via the hierarchical model in Section 6.11 closes this limitation: pooling 71 Bruchovsky patients shrinks the per-patient posterior in the most rank-deficient directions $\alpha_{T-,T+}$ and $\alpha_{T-,TP}$ by median factors of ~ 0.13 and ~ 0.14 respectively (86–87% reduction in posterior std). All hierarchical R-hat statistics pass the clinical-grade $\hat{R} < 1.10$ threshold under the non-centered parameterization.

2. **Noise model.** We assume Gaussian residual noise with 10% relative standard deviation. Real PSA assays have additive + multiplicative + outlier components.

3. **Synthetic-data validation.** Section 5's MCMC validation uses synthetic data; the actual posterior on real Brady-Nicholls 2020 cohort fits may differ.

4. **Schedule generality.** Section 4 tests three schedules. Other schedules may give different rank in principle.

7.3 Implications for clinical adaptive therapy

1. **Stop attempting full per-patient L-V parameter fits from PSA alone.** The math (Sections 2–3) and the empirical convergence (Brady-Nicholls / Strobl / Gallagher) both say at most 1–3 effective parameters are recoverable.

2. **Prefer MCMC over FIM-pseudoinverse for posterior characterization** when working in the rank-deficient regime (Section 5).

3. **Multi-modal observation channels improve precision but do NOT close the rank gap** on the 3-pop K-shift model (see Section 6.10). They are useful for tightening the identifiable subspace, not for delivering per-patient θ recovery. Full per-patient identifiability requires *either* model simplification *or* cohort-level pooling. There is no observation-channel shortcut.

4. **Posterior-aware control may not always change the policy choice**, but knowing *when* it does is a question this paper begins to address.

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Appendix A — Proof of Theorem 1

We prove a precise version of the Theorem stated informally in Section 2: under PSA-only observation of (1)–(2), the parameters α and β admit a one-parameter family of values that produce identical observation trajectories to leading order in perturbation.

A.1 Setup

Let $\theta_0 = (r_S, r_R, \alpha_0, \beta_0, K, d) \in \mathbb{R}^6_{>0}$ be a nominal parameter vector with $\alpha_0, \beta_0 \in (0,1)$ (the AT-relevant coexistence regime). Let $X(t;\theta) = (S(t;\theta), R(t;\theta))^\top$ denote the solution of (1) with control $u(t)$ from a fixed admissible class, initial condition $X(0) = X_0 = (S_0, R_0)^\top$, and parameters θ.

We consider perturbations $\theta_0 + s\,v$ along an arbitrary direction $v \in \mathbb{R}^6$, $s \in \mathbb{R}$ small. The PSA observable from (2) is, to leading order,

$$y(t;\theta) = \rho\,(S(t;\theta) + \gamma R(t;\theta)) - \phi P(t;\theta),$$

which is a linear function of X for fixed P, and P itself is an integral of X over $[0,t]$. Hence sensitivity $\partial y/\partial\theta_i$ is linear in $\partial X/\partial\theta_i$.

A.2 Sensitivity equations

The first-order parameter sensitivities $X_{\theta_i}(t) := \partial X(t;\theta)/\partial\theta_i|_{\theta=\theta_0}$ satisfy the variational system

$$\dot{X}_{\theta_i}(t) = J(t)\,X_{\theta_i}(t) + g_{\theta_i}(t), \qquad X_{\theta_i}(0) = 0,$$

where $J(t) = \partial f/\partial X|_{X(t;\theta_0)}$ is the time-varying Jacobian along the nominal trajectory, and $g_{\theta_i}(t) = \partial f/\partial\theta_i|_{\theta_0,X(t;\theta_0)}$ is the parameter-source term. For f given by the right-hand side of (1):

$$g_\alpha(t) = \begin{pmatrix} -r_S S(t)R(t)/K \\ 0 \end{pmatrix}, \qquad g_\beta(t) = \begin{pmatrix} 0 \\ -r_R R(t)S(t)/K \end{pmatrix}.$$

A.3 Key observation: parallel source terms after PSA projection

Define the PSA-weighted state vector $w := (1,\gamma)^\top$ so that $y = \rho\,w^\top X - \phi P$. The PSA-projected sensitivity to α is

$$\frac{\partial y(t)}{\partial\alpha} = \rho\,w^\top X_\alpha(t) - \phi\,P_\alpha(t),$$

where $P_\alpha(t) = \int_0^t [\rho\,w^\top X_\alpha(\tau) - \phi P_\alpha(\tau)]\,d\tau$. Since P_α is a linear filter of $w^\top X_\alpha$, the question of whether $\partial y/\partial\alpha$ and $\partial y/\partial\beta$ are linearly dependent reduces to whether $w^\top X_\alpha(t)$ and $w^\top X_\beta(t)$ are linearly dependent for all t.

A.4 The α-β degeneracy in the high-S, low-R regime

Consider the AT-relevant regime where $R(t) \ll S(t)$ for all t in the trajectory's PSA-observable window. In this regime, $g_\alpha(t)$ and $g_\beta(t)$ have the structure $g_\alpha(t) = c_\alpha(t)\,e_S$ with $c_\alpha(t) = -r_S S(t)R(t)/K$ and $e_S = (1,0)^\top$; $g_\beta(t) = c_\beta(t)\,e_R$ with $c_\beta(t) = -r_R R(t)S(t)/K$ and $e_R = (0,1)^\top$. Note that $c_\alpha(t)/c_\beta(t) = r_S/r_R$ is a *constant* along the trajectory.

A.5 Conclusion

Define c as the ratio of PSA-projected sensitivity convolutions at any sample time. Then for any $s \in \mathbb{R}$ small,

$$y(t;\theta_0 + s(c\,e_\alpha - e_\beta)) = y(t;\theta_0) + O(s^2) + O(s\gamma) + O(s\,R(t)/S(t)).$$

Hence the one-parameter family $\theta(s) := \theta_0 + s(c\,e_\alpha - e_\beta)$ produces identical PSA observations to leading order. The corresponding linear direction in (α,β)-space is the eigenvector of the FIM with eigenvalue 0 (or, in the perturbed case, the smallest eigenvalue), confirming the rank-1 (over α,β) result of Section 2. $\square$

Appendix B — Reproducibility

All numerical results trace to experiment scripts in `src/experiments/` of the open-source repository at `aplab-dev/wp1-fim-identifiability`. Each script is deterministic given a seed; output files include git SHA and ISO date in the filename. The full reproducibility table is included in the markdown source (`writeups/preprints/WP1_FIM_methods_note.md`).